Project Proposal

1. Project Title

Atica Canteen Management System (CMS)

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2. Project Team

S/N	Index Number	Name	Role
1	5221040060	NTIAMOAH RAZAK	Project Manager, Lead Developer
2	5221040057	SHAMSUDEEN HUMUHAIRI	UI/UX Designer & Documentation
3	5221040060	PAUL OSEI APRAKU	Researcher / Analyst

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3. Project Abstract

The Canteen Management System (CMS) is a streamlined, digital-first solution designed for schools to replace paper-based collection, prepayment, and owing tracking systems. It provides Administrators and Teachers with role-based dashboards to manage daily collections, student balances, expenses, revenue, and profit insights. The system delivers consistent, secure, and auditable records while enabling detailed analytics and reducing administrative workload. With offline-capable data capture and easy integration paths for student information systems, the CMS ensures operational resilience and transparency in environments with varying infrastructure readiness.

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4. Introduction

Project Background

School canteens rely heavily on manual processes—paper logs, notebooks, or improvised spreadsheets—to track meal collections, student debts, and prepayments. This often results in inconsistent records, reconciliation delays, and limited visibility for decision-makers. Modern research on school micro-finance digitization shows that digital ledgers, cashless/prepaid systems, and structured reporting significantly enhance transparency, accuracy, and trust.

Project Goals

- Digitize all collection and financial workflows in a secure, auditable manner.
- Provide real-time visibility of student balances and canteen financials.
- Ensure easy adoption with offline support, simple UX, and integration flexibility.
- Reduce administrative effort and improve school-wide accountability.

Project Scope

Inclusions

- Role-based access for Administrators and Teachers.
- Student account management: prepayments, top-ups, owings.
- Collection session recording with offline capability.
- Financial tracking: expenses, revenues, and profit analysis.
- Multi-dashboard analytics and exportable reports.
- Integration-ready structure for SIS, SSO, and payments.

Exclusions

- Direct food ordering by students.
- Loyalty/rewards programs.
- Automated supplier procurement.

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5. Target Audience

Primary Users

- School Administrators overseeing finances and policy compliance.
- Teachers managing class-level collection sessions.

Secondary Users

- Finance officers/auditors.
- Students (indirectly, via statements or balance notifications).

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6. System Requirements

Functional Requirements

- User registration/login with RBAC.
- Student account management: top-ups, deductions, owings, aging.
- Offline-capable collection sessions with sync/reconciliation.

- Expense, revenue, and profit tracking modules.
- Reporting and analytics dashboards (daily/weekly/monthly).
- CSV/Excel/PDF export for finance and audit purposes.
- Optional payment gateway integration for top-ups.

Non-Functional Requirements

- Scalability for schools of different sizes.
- High availability and automatic syncing after offline use.
- Secure storage and encrypted transmission of sensitive data.
- WCAG 2.1 AA accessibility.
- Mobile-responsive and low-bandwidth friendly.

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7. System Architecture

The CMS adopts a modular, three-tier architecture similar to modern enterprise school systems:

Frontend

- React.js / Next.js
- Tailwind CSS and ShadCN/UI for consistency and usability
- PWA capabilities (offline data capture via IndexedDB)

Backend

- Node.js with NestJS or Express.js
- REST or GraphQL APIs
- Audit logging and RBAC middleware

Database

- PostgreSQL for transactional records
- Redis for caching and offline sync queues

Integrations

- Student Information System (SIS) sync
- Payment gateway (Stripe/Paystack) optional
- CSV/Excel bulk import/export for deployment flexibility

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8. Development Methodology

Agile Methodology (Scrum)

The project will follow Agile due to its adaptability and focus on iterative delivery. Each sprint delivers usable components, enabling stakeholder review and early feedback. This approach improves quality, reduces adoption risk, and ensures alignment with school workflow realities.

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9. Project Schedule

Milestone	Duration	Deadline
Requirement Gathering	2 Weeks	Week 2
System Design	2 Weeks	Week 4
Frontend Development	3 Weeks	Week 7
Backend Development	4 Weeks	Week 11
Integration & Testing	2 Weeks	Week 13
Deployment & Handover	1 Week	Week 14

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10. Testing & Quality Assurance

- **Unit Testing:** Validate core components and logic.
- **Integration Testing:** Ensure smooth communication across modules.
- **System Testing:** Validate end-to-end correctness of features.
- **User Acceptance Testing:** Collect feedback from teachers and administrators.

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11. Resources

Hardware

- Laptops/desktops with stable internet (intermittent acceptable due to offline mode).

Software

- VS Code, PostgreSQL, Postman, Browser DevTools
- GitHub for version control
- Paystack/Stripe SDK (optional)

Personnel

- Developers, QA testers, UI/UX designer
- Documentation officer
- School IT liaison

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12. Project Deliverables

- Fully functional Canteen Management System
- Administrator and Teacher dashboards
- Training manuals and technical documentation
- Demo video and presentation slides
- Deployment guide and SOPs for school workflows

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13. Success Criteria

- Accurate tracking of prepayments, owings, and collection sessions
- Reduction in reconciliation time and paper dependency
- Consistent, audit-ready records with minimal errors
- Positive user feedback during UAT
- Completion of milestones within scope and timeline

References

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- GDPR. (2016). *General Data Protection Regulation*. Official Journal of the EU.
- Research on cashless school finance systems, offline-first PWAs, and audit-centric ledger design (additional citations available in appendix).